

Assessment of vehicle dynamics using high-performance driving simulator

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Abstract – The aim of this work is to establish a practical evaluation method to measure the performances of driving simulators and the motion cueing algorithms in terms of relative or absolute fidelity. To achieve this objective, we investigate two driving experiences for lateral vehicle dynamics using a high-performance driving simulator. One scenario involves normal freeway driving tests with slalom maneuvers, while the other entails driving at the tire-adherent limit condition on a handling track. We compare the performance of motion cueing algorithms based on classical filter (CF-MCA) and model predictive control (MPC-MCA) using the method developed in this study which is consistent with the ratings given by professional drivers. Thanks to this work, a vehicle handling assessment on high lateral dynamics has been achieved on ROADS (Renault Operational Advanced Driving Simulator) and is applicable for all vehicles' project development.

Keywords: MCA performance, driving simulation for vehicle dynamics assessment, tilt velocity threshold.

1. Introduction

Driving simulators have a wide range of applications: driver training, optimizing race car design, such as the F1 driving simulators at Renault Sport and McLaren in the UK, road infrastructure design (e.g., Tongji University driving simulator), vehicle development with the driver in the loop, testing critical situations without exposing drivers to real danger, etc. In recent years, with the growing development of AD/ADAS and electric vehicles, some car manufacturers have developed high-performance driving simulators (e.g., Daimler, Nissan, BMW, Renault, etc.). The main aim of these high-performance driving simulators is to reduce the gap in motion perception between the driving simulator and the real car, thereby improving the fidelity of the driving simulation and making the subjective evaluation results more representative of the physical tests. Thus, by using an advanced driving simulator, we can carry out subjective evaluations using digital models upstream of the project. As described in the literature (Kharrazi et al., 2019), driving simulators can be used for iterative testing at an early stage to better gather subjective understanding of various vehicle designs. This will enable vehicle designs to be evaluated and refined before moving on to real-world testing, thereby reducing development time and improving design quality.

In the literature, we found a few articles dealing with the subjective assessment of vehicle dynamics using a driving simulator (Fischer et al., 2010; Berthoz et al., 2013; Kharrazi et al., 2019; Lamprecht et al., 2022), but these were generally limited to a level of lateral acceleration on freeways between 2 and 3 m/s². Studies investigating high lateral dynamics in a motion-based driving simulator (Fang et al., 2020) are still rare. To fill this gap, the present study investigates the subjective assessment of high lateral dynamics based on different scenarios ranging from highway driving experiences with slalom to handling tracks. The aim of this work is to establish a practical evaluation method in terms of relative or absolute fidelity for Renault's high performance driving simulator ROADS.

The performance of Renault's driving simulator, ROADS, is presented in Table 1.

Table 1: Performance of ROADS

Rail length (m)		Rail velocity (m/s)		Rail accel. (m/s ²)		Pitch, Roll, Yaw table angle (°)	
X	Y	X	Y	X	Y	Pitch/Roll	Yaw table
±7.5	±7.5	±9	±9	±10	±10	±23	±165

2. Description of scenarios

To evaluate the behaviors of vehicle lateral dynamics on the ROADS driving simulator, we developed several scenarios. In this paper, the two most important scenarios are described:

2.1 N104 scenario

This scenario is based on the French national road named N104. N104 is a fast-way road with a maximum driving speed of 110 km/h. To evaluate higher levels of lateral acceleration of the vehicle, two series of slalom maneuvers are added. The first one is designed with a sinusoid of 0.2 Hz at 70 km/h, and the second one with 0.2-0.3 Hz at 70 km/h. For this test, traffic is also simulated to provide a more immersive experience.

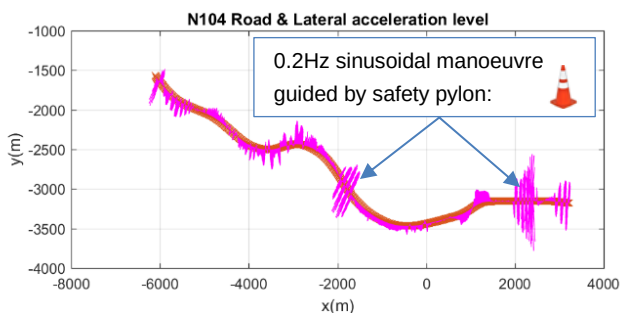


Figure 1: N104 road

2.2 CTA1 scenario

The CTA1 scenario corresponds to our vehicle handling evaluation track on the proving ground. It contains different road curvatures with radii varying from 35m to 250m. For this driving test, the driver is asked to drive the car almost at the limit of tire adherence. The maximal lateral acceleration approaches 1g in every bend.

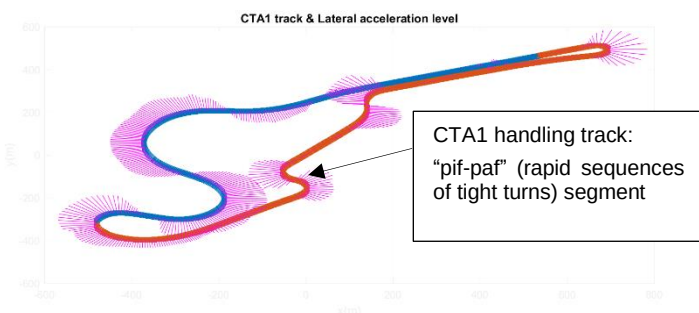


Figure 2: CTA1 handling track

3. Potential workspace of a driving simulator

The potential workspace or more precisely the potential reproduction area of acceleration in a driving simulator is defined by the acceleration level and its maximal duration that can be simulated. We

will establish this potential operating area considering human motion perception thresholds. We refer to this potential operating zone as a " γ -T chart" for simplicity.

In a dark room, the perceived linear motion threshold is 0.05m/s^2 for sinusoidal excitation (Benson et al., 1986). In a driving simulator with high dynamic scenarios, this threshold could increase to 0.1m/s^2 , or even more (cf. table 2). The perceived tilt velocity in a driving simulator generally ranges from $3^\circ/\text{s}$ to $6^\circ/\text{s}$, depending on the current acceleration amplitude reproduced by rail and hexapod. Our experience in a high dynamic scenario with high rail motion shows that a tilt velocity of $4^\circ/\text{s}$, or even 5 - $6^\circ/\text{s}$ in some instances of high peak acceleration, can be accepted by most people. Therefore, a linear acceleration of 0.1m/s^2 and a tilt velocity of $4^\circ/\text{s}$ are used to construct the γ -T map.

Table 2: Motion perception thresholds

Axis	Linear motion (m/s^2)		Angular motion ($^\circ/\text{s}$)	
	In darkness (Benson86)	Dynamic scene (Reid85)	In darkness (Benson86)	Dynamic scene (Reid85)
x	0.063	0.17	2.04	3.0
y	0.057	0.17	2.07	3.6
z	0.154	0.28	1.5	2.6

To reproduce constant acceleration motion, the cockpit is generally accelerated constantly for a duration of T seconds, followed by a long braking phase to stop at the end of the maximal rail position. The braking phase is controlled with a deceleration level below the human perception threshold. Figure 3 illustrates an example of ROADS X or Y rail performance for reproducing 1g acceleration using $U_{\text{brake}} = 0.1\text{m/s}^2$. The reproduced period ($\approx 0.18\text{s}$) is very short for this example. By employing an opposite motion strategy, named "back-go strategy" to generate negative velocity (Fang, 2010, see Fig. 4), the simulated duration of acceleration can be doubled. However, the second case is not applicable for an active driving test, except for some event-triggered tests where the car's trajectory is always the same and the vehicle speed is repeated in each test. To increase the simulated acceleration duration, we can also use tilt coordination technique to achieve higher braking motion without being perceived by the driver, thus significantly extending the reproduced duration.

Figure 4 illustrates the principle of the strategy using tilt-coordination and pre-velocity technique. The principle of the "back-go strategy" with tilt is mainly used in MPC-MCA or optimal-MCA (Venrooij et al., 2016).

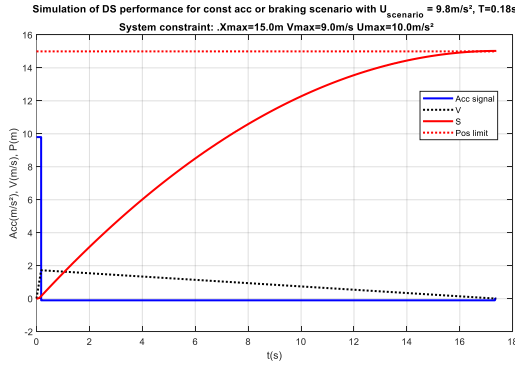


Figure 3: 1g reproduction using only one rail

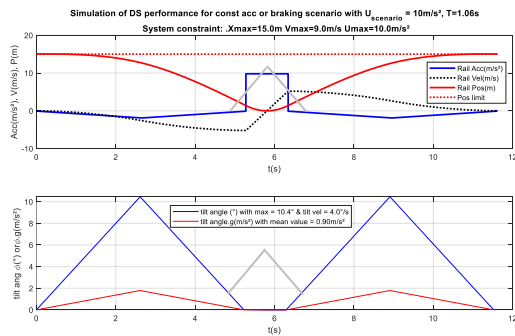


Figure 4: Back-go strategy for getting more acc. duration

Note that in this example, the tilt-coordination technique is only used for the braking phase. We can slightly improve the motion cueing results if it is also used in the acceleration reproduction phase (see the small triangle around $t=6s$).

In terms of pure tilt motion, it can provide a sustained acceleration feeling but is limited by the motion perception threshold of the tilt velocity and the maximum tilt angle. Using a tilt velocity of $4^\circ/s$, we can define its operating zone in the γ -T map (see Fig. 5: tilt specific force zone). Without the rail motion, the reproduced tilt specific force has a rate limitation on acceleration or jerk limitation.

Finally, we can apply the full tilt-coordination technique and couple X and Y rail motions to define the maximal γ -T operating zone. As it depends strongly on the performance of developed MCA, the tilt-coordination zone is arbitrarily depicted by a dotted line (see Fig. 5: tilt-coordination technique).

Fig. 5 presents the ROADS driving simulator's γ -T chart and some handling use-cases. The N104 scenario refers to high-speed Lane/Change. The CTA1 scenario represents the furthest zone to enter the ROADS operating workspace. In order to incorporate the CTA1 operating zone into the exploitable γ -T workspace of ROADS, the reference acceleration to be reproduced is modified by using two scaling factors:

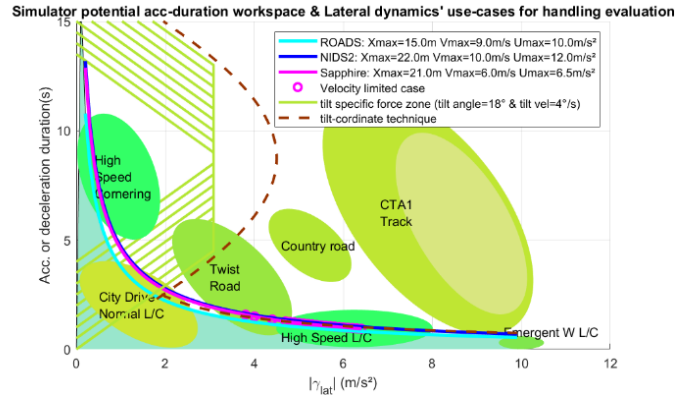
$$\gamma_{lat ref} = K_{LP} \cdot (\gamma'_{veh} - \gamma_{lat HP}) + K_{HP} \cdot \gamma_{lat HP}$$

$$\text{with } \gamma'_{lat veh} = \gamma_{lat veh} - \phi_{veh} \cdot g$$

$$\gamma_{lat HP} = HP \text{ filter}(\gamma'_{lat veh})$$

which is judged much better than a constant scaling factor for CTA1 scenario.

In the reference signal's formula, γ denotes the acceleration, ϕ_{veh} is the vehicle natural roll angle.

Figure 5: Driving simulator operation zone and use-cases working zone in γ -T chart

As depicted in Fig. 5, the lowpass scaling factor should ideally be around 0.3-0.4 for the CTA1 scenario. It's important to note that a real-time filter with minimal data buffering latency introduces some phase distortion to the filtered signal. Specifically, the phase of high-frequency signals is advanced while low-frequency signals present a delay (Fang et al., 2020). If the lowpass and highpass scaling factors have the same value, the resultant reference signal's phase aligns perfectly with the original signal. However, in some use-cases, drivers prefer more feedback from high-frequency signals to better understand the vehicle's behavior. Unfortunately, when using two different scaling factors, the phase of the resultant signal is distorted compared to the original. These scaling factors need to be tuned based on the performance of the motion cueing algorithm and the driver's tolerance for false cues and phase distortion.

4. MCA description

To evaluate MCA performance, we compare two motion cueing algorithms: one based on classical filter technique, a widely used MCA, and another, our internal Model Predictive Control based MCA (MPC-MCA).

To ensure consistent motion perception across lateral acceleration, roll, and yaw velocity motions, all three signals are scaled using the same low-frequency scaling factor, K_{LP} , and high-frequency scaling factor, K_{HP} . The cutoff frequency of the applied filter is also the same for these three motions.

Table 3: 2nd order lowpass (LP) & highpass (HP) filter

Filter Type	Filter transfer function: H(s)	Pulsation ω_n , or cut-off frequency f_c
2 nd LP	$\frac{\omega_n^2}{s^2 + 2\xi \cdot \omega_n \cdot s + \omega_n^2}$	$\omega_n = (2\pi f_c)$
2 nd HP	$\frac{s^2}{s^2 + 2\xi \cdot \omega_n \cdot s + \omega_n^2}$	$\omega_n = (2\pi f_c)$

Table 4: Scaling parameters & HP filter parameters

CTA1 γ_{lat} , ω_{yaw} & ω_{roll} ($^\circ/s$)	HP f_c (Hz)	damping coef. ξ	K_{LP}	K_{HP}
MPC-MCA	0.15	0.85	0.5	0.9
CF-MCA	0.15	0.55	0.12	0.6

4.1 Classical filter-based MCA

The parameters of Classical Filter-based MCA (CF-MCA) are tuned to maximize the linear correlation coefficient with the original signal for the HP signal while keeping the HP signal's amplitude as high as possible (refer to Table 4 for filter details). The specific force is reproduced by the linear motion generated by rail or hexapod and by the tilt specific force provided by hexapod thanks to the tilt angle and the gravity force.

For the linear motion, we have:

$$\gamma_{lat \text{ lin ref}} = K_{LP} \cdot (\gamma'_{lat \text{ veh}} - \gamma_{lat \text{ lin HP}}) + K_{HP} \cdot \gamma_{lat \text{ lin HP}}$$

The scaling factor K_{LP} is mainly determined by the workspace.

The scaling factor K_{HP} must strike a balance between producing sufficient amplitude of high frequency motion while avoiding significant false cues due to the HP filter behaviours such as “backlash” effect (Reymond et al., 2000) and phase distortion. The use of incompatible K_{LP} and K_{HP} scaling factors could cause a very disturbing perception of motion for some frequency signal like the motion at “pif-paf” segment in the CTA1 scenario.

Regarding the tilt motion, it is tuned by:

$$\phi_{simulator} = \frac{\gamma_{lat \text{ veh}} - \gamma_{lat \text{ lin ref}}}{k_{tilt} \cdot g} + k_\phi \phi_{veh}$$

$$\text{with } \left| \frac{d(\phi_{simulator} - \phi_{veh})}{dt} \right| \begin{cases} < 5.5^\circ/s \text{ for CTA1} \\ < 3^\circ/s \text{ for N104} \end{cases}$$

4.2 MPC based MCA

The MPC-based MCA, an internal software called ROMA (Renault Optimal Motion-cueing Algorithm), employs a variable time-step algorithm to reduce CPU time. Operating at 200Hz on a Concurrent Real-Time system, it distributes a predictive horizon of about 10s over 50 steps (see Fig. 6). Nominal tilt velocity control is set to 4°/s, but adaptable to 5 or 6°/s under high solicitation and rail acceleration conditions. The threshold of tilt velocity adapted to the rail acceleration level was reported by (Chapron et al. 2007).

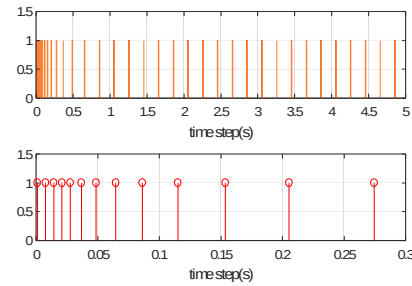


Figure 6: Variable time step of MPC-MCA

The average computational time is approximately 2ms using a single-core CPU.

5. MCA performance measurement

5.1 False cues classification

To analyze the false cues, Kolff et al. (2022) propose to classify the false cues into 4 types which can be mathematically expressed by:

Table 5: Classification of false cues according to Kolff et al.

Type of false cues	Logical expression
1 Scale mismatched	$ k_{eff} \cdot \gamma_{veh} - \gamma_{MCA} > 0.3 k_{eff} \cdot \gamma_{veh} $ & $ k_{eff} \cdot \gamma_{veh} - \gamma_{MCA} > u_{thd}$
2 False detection	$ k_{eff} \cdot \gamma_{veh} > u_{thd}$ & $ \gamma_{MCA} < u_{thd}$ Vehicle moving, MCA immobile
3 False detection	$ k_{eff} \cdot \gamma_{veh} < u_{thd}$ & $ \gamma_{MCA} > u_{thd}$ MCA moving, Vehicle immobile
4 Opposite motion	$ k_{eff} \cdot \gamma_{veh} > u_{thd}$ & $ \gamma_{MCA} > u_{thd}$ & $sign(\gamma_{veh}) = -sign(\gamma_{MCA})$

Fig. 7 illustrates the domain of type 2 false cues, similar to type 3.

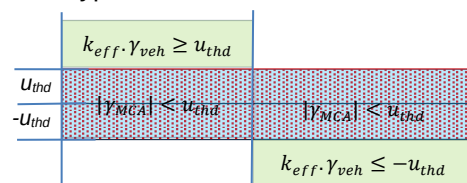


Figure 7: Domain of type 2 of false cues

However, some points of type 2 false cues (Fig. 8) should not be classified as false cues, as they move following some trajectory. Thus, it's preferred not to consider this domain as part of the false cues.

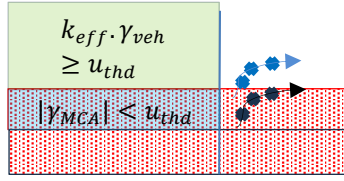


Figure 8: Illustration of removal false cues

Note that the type 1 of scale false cues is not completely independent of type 2 and 3. The original classification method is slightly modified as follows:

Table 6: Classification of false cues

Type of false cues	Logical expression
2	$ k_{eff} \cdot \gamma_{veh} > u_{thd} \ \& \ \gamma_{MCA} < u_{thd}$ $\& \ k_{eff} \cdot \gamma_{veh} - \gamma_{MCA} > u_{thd}$
3	$ k_{eff} \cdot \gamma_{veh} < u_{thd} \ \& \ \gamma_{MCA} > u_{thd}$ $\& \ k_{eff} \cdot \gamma_{veh} - \gamma_{MCA} > u_{thd}$
4	$ k_{eff} \cdot \gamma_{veh} > u_{thd} \ \& \ \gamma_{MCA} > u_{thd}$ $\& \ sign(\gamma_{veh}) = -sign(\gamma_{MCA})$
1	$ k_{eff} \cdot \gamma_{veh} - \gamma_{MCA} > 0.3 k_{eff} \cdot \gamma_{veh} $ $\& \ k_{eff} \cdot \gamma_{veh} - \gamma_{MCA} > u_{thd}$ $\& \ \sim (type\ 2 type\ 3 type\ 4)$

The original proposed statistic value EAP (error area per unit time) is replaced by a mean absolute motion cues' error which is more significant in MCA score evaluation.

5.2 Other statistic indices

To provide more comprehensive statistical data, we introduced three additional macroscopic indices to create a more representative global score. These additional indices are summarized in table 7.

Table 7: statistic indices used in global score

Additional index	Formula
Correlation 1 - R^2 R^2 : coefficient of determination	$R = \frac{\sum(\gamma_{veh\ i} - \bar{\gamma}_{veh}) \cdot (\gamma_{MCA\ i} - \bar{\gamma}_{MCA})}{\sqrt{\sum(\gamma_{veh\ i} - \bar{\gamma}_{veh})^2 \cdot \sum(\gamma_{MCA\ i} - \bar{\gamma}_{MCA})^2}}$ R: Pearson correlation coefficient
Correlation 2 - $R_{\%}$	$R_{\%} = \text{percentage of points}$ $\text{as } (k_{eff} \gamma_{veh\ i} - \gamma_{MCA\ i}) < u_{thd}$ $\text{with } u_{thd} = 0.2\text{m} \cdot \text{s}^{-2} \text{ for HP signal}$ $u_{thd} = 0.5\text{m} \cdot \text{s}^{-2} \text{ for all frequency signal}$
Correlation 3 - R_n	$R_n = \left(1 - \frac{\sqrt{\sum(k_{eff} \gamma_{veh\ i} - \gamma_{MCA\ i})^2}}{k_{eff} \sqrt{\sum(\gamma_{veh\ i} - \bar{\gamma}_{veh})^2}} \right) \cdot 100$

Correlation 1 is based on the Pearson coefficient to evaluate the quality of MCA. In scenarios where two scaling factors are involved in the reference signal, The correlation of motion cues between MCA and vehicle is carried out using respectively filtered LP & HP frequency signals. The linear coefficient of the model corresponds to the effective scaling factor respectively for LP and HP motions, and R^2 represents the coefficient of determination, where R is the Pearson coefficient (see Figs. 13-18). It generally provides an optimistic MCA performance score.

Correlation 2 is based on histogram results where the correct motion cues are determined by:

$$|(k_{eff} \gamma_{veh\ i} - \gamma_{MCA\ i})| < u_{thd}$$

Different motion thresholds (see table 8), u_{thd} , are used according to the vehicle dynamics and the frequency band, providing a score consistent with the subjective ratings given by professional drivers.

Table 8: Acceptable motion cues' threshold

scenario	for HP signal	for whole signal
N104	0.05m/s ²	0.2m/s ²
CTA1	0.2m/s ²	0.5m/s ²

The signal from all frequency bands is utilized, penalizing scenarios with two scaling factors. The index of correlation 2 is normally close to the sum of correct and scale false cues in the classification method.

Correlation 3 assesses the significance of any large differences between the vehicle motion cues and the reproduced motion (type 4 of false cues) using whole signal.

Using all frequency bands signal to make the correlation index can greatly penalize the MCA's performance with the strategy of multiple scaling factors due to the linear correlation model assuming only one effective scaling factor. Since we have two indices ($R_{\%}$ & R_n) obtained by considering all frequency bands' signals and two indices provided by separated LP & HP signals R^2 & R_c , the correct cues in classification method), the global score is balanced.

For the CTA1 scenario, the LP signal for comparison is filtered using $f_c=0.10\text{Hz}$ compared to a reference signal filtered by HP $f_c = 0.15\text{Hz}$. The HP signal is filtered using $f_c = 0.45\text{Hz}$. Other lower cutoff frequencies can also be used.

In the global score estimation, false cues of type 4 are considered the worst, justifying a penalty coefficient three times higher than that of type 2 or 3. The penalty coefficient for false scale cues is set at half the value of the type 2 or 3.

The score for LP or whole signal, S_{LP} , is calculated by:

$$\left[\left(K_{LP\text{eff}} + \frac{R_c + R^2 + R_{\%} + R_n - w_e(0.5R_{e1} + R_{e2} + R_{e3} + 3R_{e4})}{4} \right) \right]$$

The score for HP, S_{HP} , signal is given by:

$$\left[\left(K_{eff} + \frac{R_c + R^2 + R_{\%} + R_n - w_e(0.5R_{e1} + R_{e2} + R_{e3} + 3R_{e4})}{4} \right) \right]_{HP}$$

$$score = \frac{10[w_{LP} \cdot S_{LP} + (1 - w_{LP})S_{HP}]}{2}$$

where the contribution of score from LP signal in global score is set by:

$$w_{LP} = 0.35$$

and the weight coefficient of false cues is set by:

$$w_e = 10$$

$$R_{e,i} = R_{err\ type\ i} \Delta|\gamma_i|$$

$R_{e,i}$ represents the penalty index of each type of false cues in the global score evaluation, $R_{err\ type\ i}$ the percentage of each type of false cues, $\Delta|\gamma_i|$, the mean acc error for each type of false cues (see Figs. 13-18).

The value of the score varies from 0 to 10 to measure the representativeness of physical tests by the driving simulation (absolute fidelity).

6. Results

6.1 MCAs performance

To facilitate the MCAs' performance comparison, in this paper, the MPC-MCA data comes from a driving experience recording and the result of CF-MCA is simulated using the same input signal as the MPC-MCA.

Fig. 9 directly compares the motion cueing results between CF-MCA and MPC-MCA for the N104 scenario. The specific force provided by MPC-MCA closely matches the vehicle's lateral acceleration on a 1:1 scale, while CF-MCA provides a motion cueing result scaled at 0.8:1 with a low linear correlation coefficient ($R^2=0.81$, see table 10). Both MCAs ensure tilt velocities below 3°/s.

Figs. 10-11 compare the motion cueing results of MPC-MCA and CF-MCA for the CTA1 scenario. A substantial difference is observed in the amplitude of the reproduced lowpass motion signal between CF-MCA and MPC-MCA.

Fig. 12 depicts the roll velocity of the MPC-MCA as well as the tilt velocity (deviation from the scaled roll velocity of the vehicle).

To produce more tilt specific force during moments of high lateral acceleration, the tilt velocity exceeds the restricted nominal value of 4°/s in MCA. While professional drivers have tested the scenario and not reported issues, future iterations of MCA should aim to reduce these high tilt velocities.

Figs. 13-18 present classification results of motion cues for both MCAs.

For the N104 scenario (Figs. 13-14), significant differences in the indicators of false cues between CF-MCA and MPC-MCA are observed for signals of all frequency bands. Note that high frequency motion cues ($f_{c\ HP} = 0.45\text{Hz}$) are effectively reproduced by both MCAs with scaling factors exceeding 0.95 and $R^2 > 0.95$ (see tables 9-10).

For the CTA1 scenario (Figs. 15-18), MPC-MCA achieves an effective scaling factor three times higher than CF-MCA for low-frequency signals (Figs 15-16), but with higher index of false cues due to increased motion stimuli reproduction. Feedback from drivers underscores the importance of low-frequency motion amplitude in driving experience. Similarly, the effective scaling factor of high-frequency signals for MPC-MCA surpasses that of CF-MCA with 35% more gain (see Figs 17-18). Both MCAs achieve >90% correctness of motion cues.

We present four examples of histograms to illustrate the distribution of groups of motion error using the result of MPC-MCA. The red line indicates acceptable motion cues with an error inferior to the criterion u_{thd} .

The percentage of points inside the red rectangle is given by the first legend value, and the correlation index R_n is provided in legend 3 of the histogram figure (see Figs 19-22). If the distribution of $\Delta\gamma_i$ is a Gaussian curve, we have:

$$R_{\%} = 0.95 \ \& \ u_{thd} = \mu + 2 \cdot \sigma$$

$$\text{With } \sigma^2 = \frac{\sum(\Delta\gamma_i - \mu)^2}{N}, \mu = \frac{\sum \Delta\gamma_i}{N}$$

For example, from Fig. 20 and the result of MPC-MCA for HP signal, we have :

$$R_{\%} = 0.958, \mu = 0, 2\sigma = 0.048\text{m/s}^2$$

The 2σ statistic value is very close to that of u_{thd} ($=0.05\text{m/s}^2$).

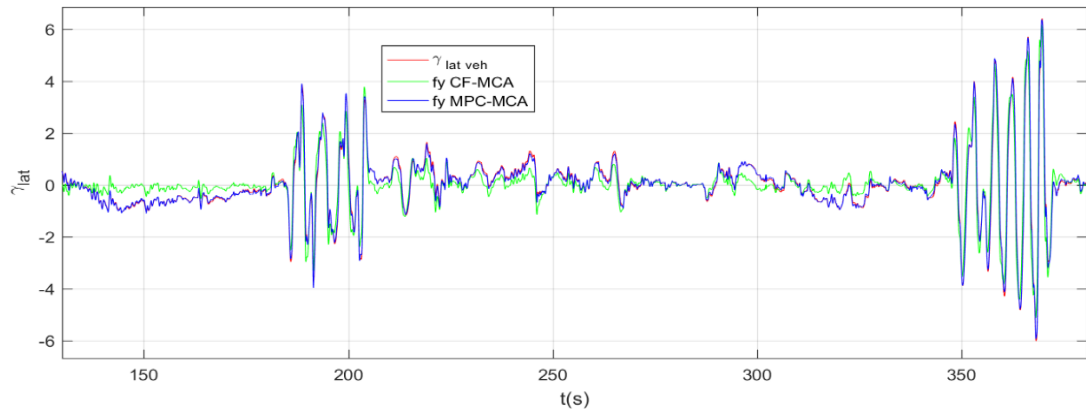


Figure 9: Comparison of motion cueing result for N104 scenario

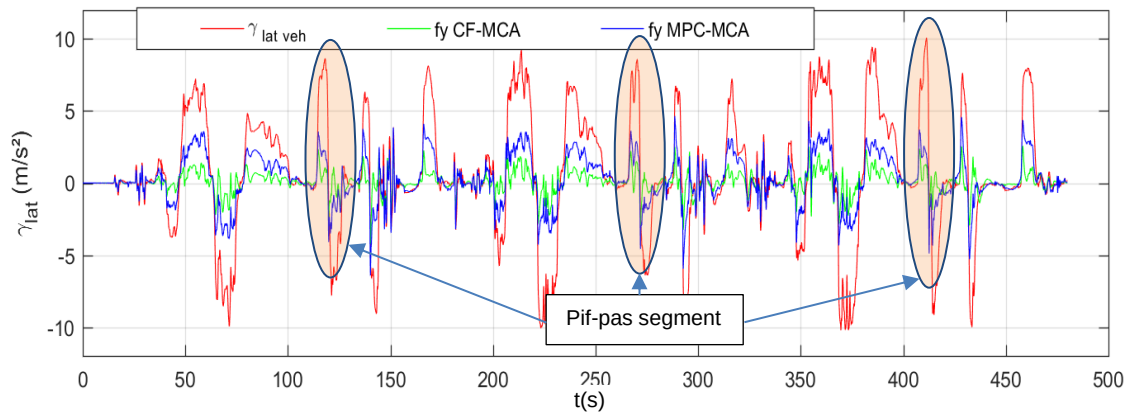


Figure 10: Comparison of motion cueing result for CTA1 scenario

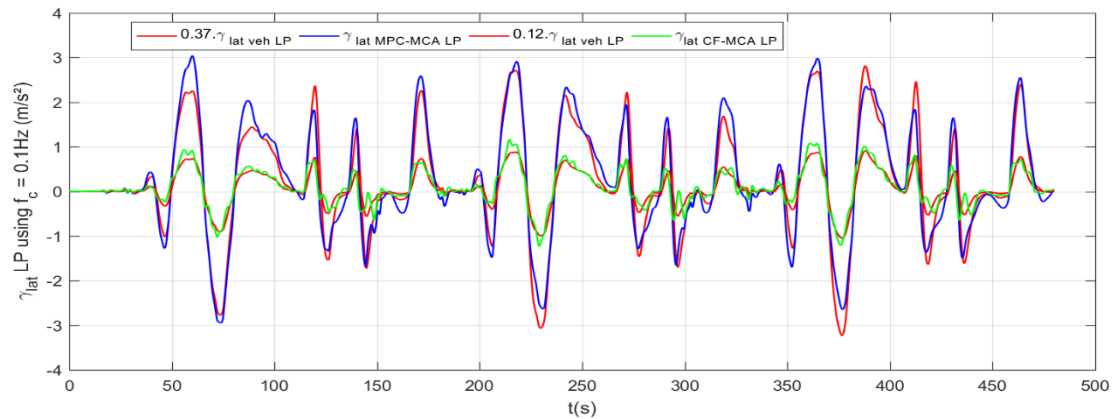
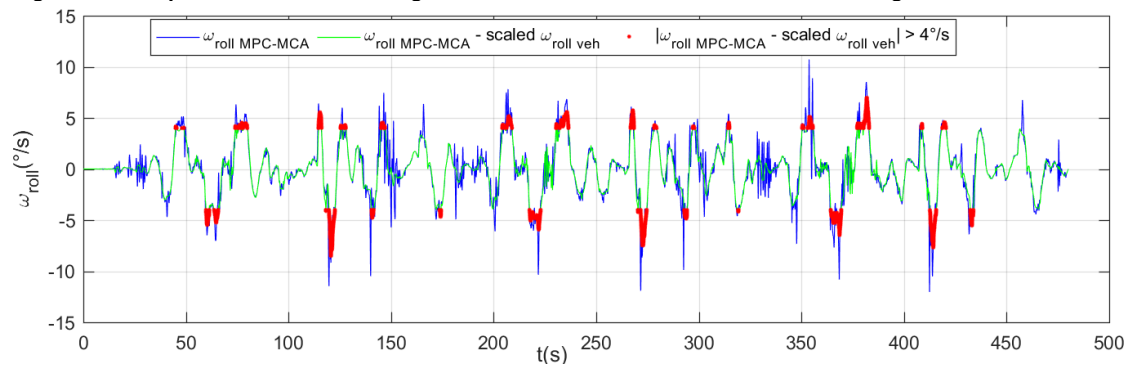


Figure 11: Comparison of motion cueing result between MPC-MCA & CF-MCA for LP signal and CTA1 scenario

Figure 12: Roll velocity in driving simulator $\omega_{roll MPC-MCA}$ and tilt roll velocity $\omega_{roll MPC-MCA} - scaled \omega_{roll veh}$

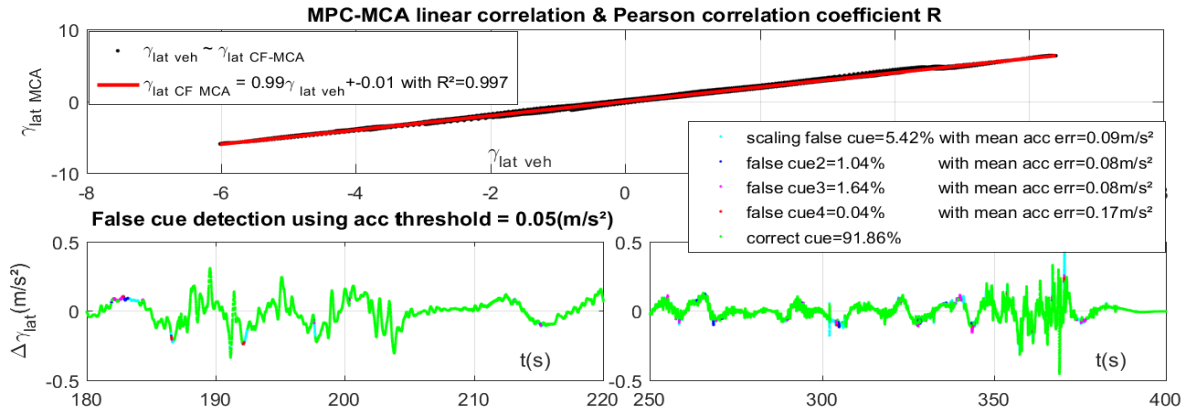


Figure 13: Classification of motion cues for MPC-MCA & N104 scenario

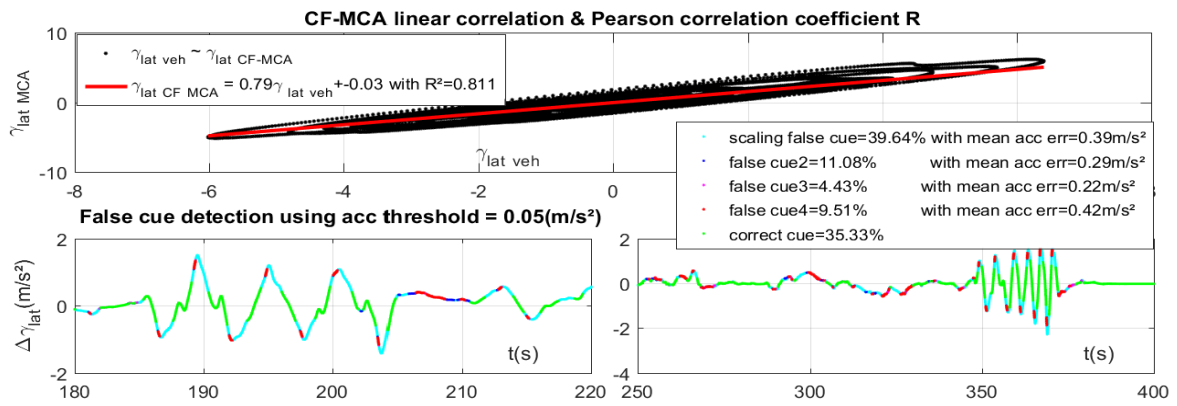


Figure 14: Classification of motion cues for CF-MCA & N104 scenario

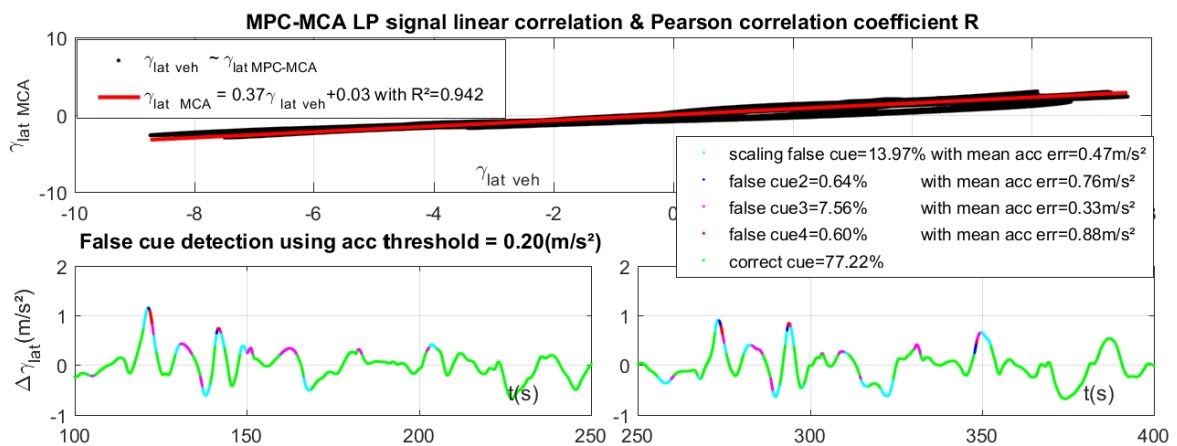


Figure 15: Classification of LP motion cues for MPC-MCA & CTA1 scenario

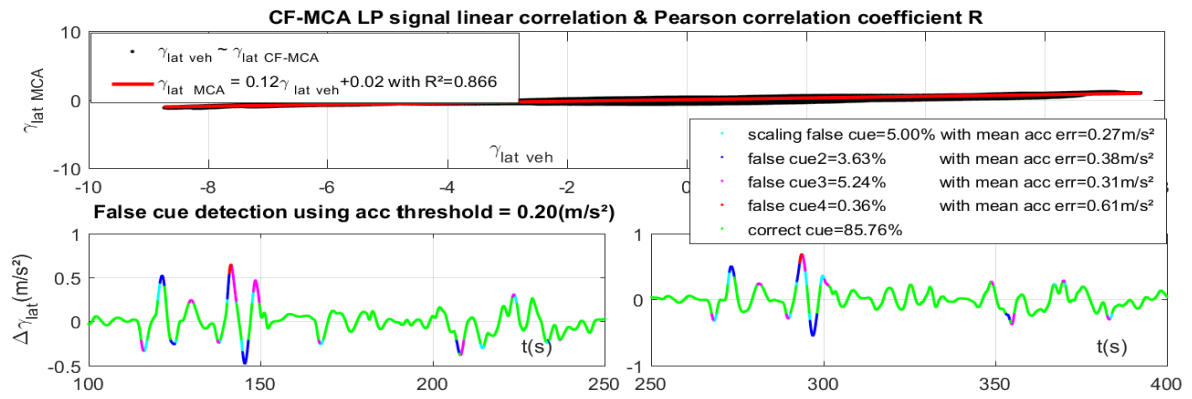
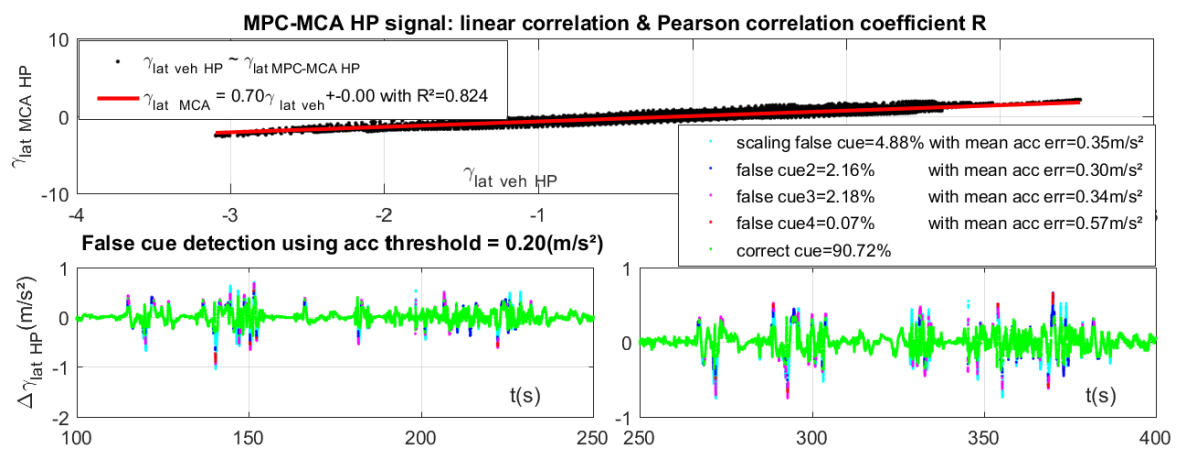
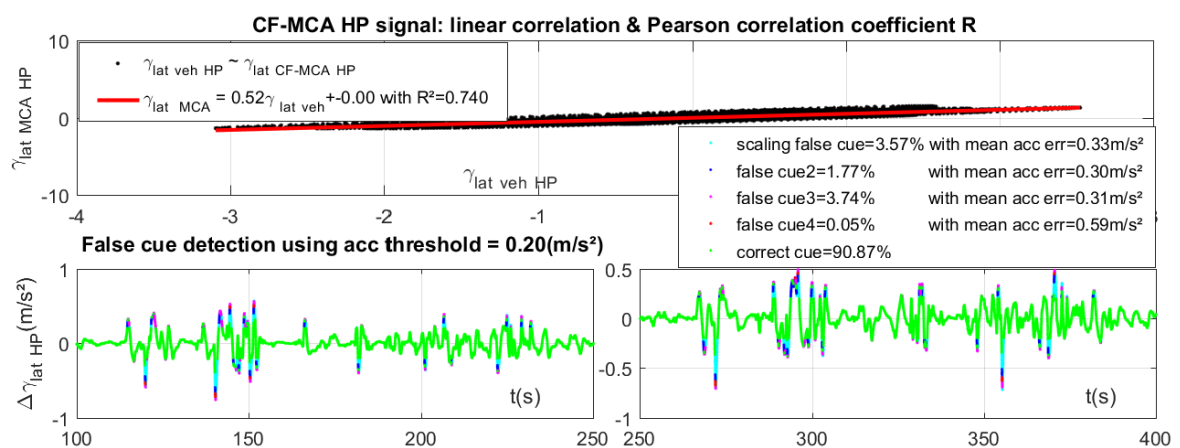


Figure 16: Classification of LP motion cues for CF-MCA & CTA1 scenario

Figure 17: Classification of motion cues for MPC-MCA & CTA1 scenario with HP signal and $f_{c\ HP} = 0.45Hz$ Figure 18: Classification of motion cues for CF-MCA & CTA1 scenario with HP signal and $f_{c\ HP} = 0.45Hz$

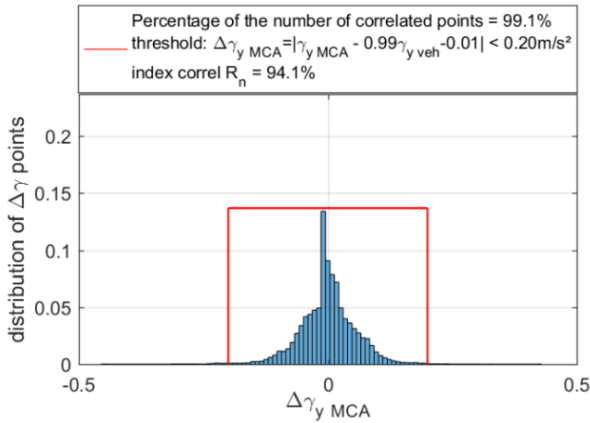


Figure 19: Histogram of whole signal for MPC-MCA & N104

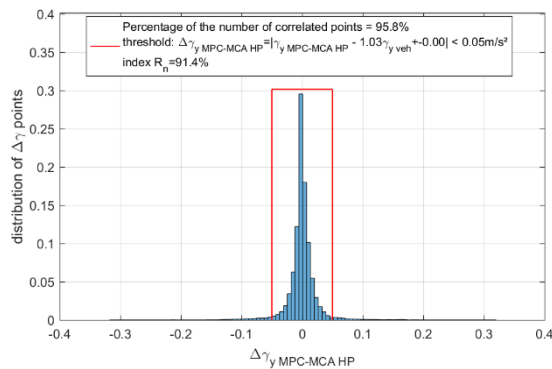


Figure 20: Histogram of HP signal for MPC-MCA & N104

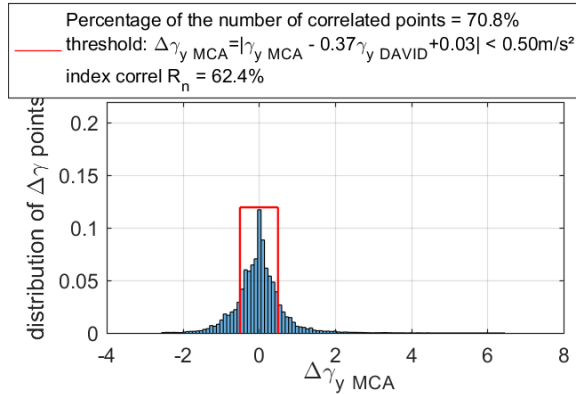


Figure 21: Histogram of whole signal for MPC-MCA & CTA1

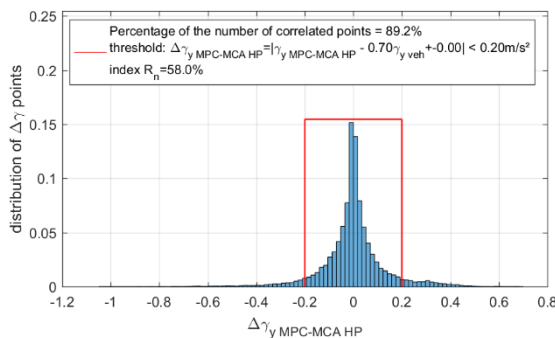


Figure 22: Histogram of HP signal for MPC-MCA & CTA1

Tables 9 to 12 summarize the correlation indices and the overall performance score of the two MCAs for the N104 and the CTA1 scenarios.

Table 9: MPC-MCA statistic indices & global score for N104 scenario

N104 MPC-MCA	Correlation index for whole signal	Correlation Index for HP signal
Correct cue (classification) $u_{thd} = 0.05m/s^2$	$R_c = 0.92$ $\sum w_{ei} R_{ei} = -0.05$	$R_{c HP} = 0.99$ $\sum w_{ei} R_{ei HP} = -0.01$
Correlation 1	$K_{eff} = 0.99$ $R^2 = 1.0$	$K_{eff HP} = 1.0$ $R_{HP}^2 = 0.99$
Correlation 2	$R_{\%} = 0.99$ using $u_{thd} = 0.20m/s^2$	$R_{\% HP} = 0.96$ using $u_{thd} = 0.05m/s^2$
Correlation 3	$R_n = 0.94$	$R_{n HP} = 0.91$
Score = 9,8		

Table 10: CF-MCA statistic indices & global score for N104 scenario

N104 CF-MCA	Correlation index for whole signal	Correlation Index for HP signal
Correct cue (classification) $u_{thd} = 0.05m/s^2$	$R_c = 0.35$ $\sum w_{ei} R_{ei} = -2$	$R_{c HP} = 0.93$ $\sum w_{ei} R_{ei HP} = -0.10$
Correlation 1	$K_{eff} = 0.79$ $R^2 = 0.81$	$K_{eff HP} = 0.95$ $R_{HP}^2 = 0.95$
Correlation 2	$R_{\%} = 0.51$ using $u_{thd} = 0.20m/s^2$	$R_{\% HP} = 0.89$ using $u_{thd} = 0.05m/s^2$
Correlation 3	$R_n = 0.57$	$R_{n HP} = 0.77$
Score = 7.4		

Table 11: MPC-MCA statistic indices & global score for CTA1

CTA1 MPC-MCA	Correlation index for LP or whole signal	Correlation Index for HP signal
Correct cue (classification)	$R_{c LP} = 0.77$ $\sum w_{ei} R_{ei LP} = -0.79$	$R_{c HP} = 0.91$ $\sum w_{ei} R_{ei HP} = -0.24$
Correlation 1	$K_{eff LP} = 0.37$ $R_{LP}^2 = 0.94$	$K_{eff} = 0.70$ $R_{HP}^2 = 0.82$
Correlation 2	$R_{\%} = 0.71$ using $u_{thd} = 0.50m/s^2$	$R_{\%} = 0.89$ using $u_{thd} = 0.20m/s^2$
Correlation 3	$R_n = 0.62$	$R_n = 0.58$
Score = 6,3		

Table 12: MPC-MCA statistic indices & global score for CTA1

CTA1 CF-MCA	Correlation index for LP or whole signal	Correlation Index for HP signal
Correct cue (classification)	$R_{c LP} = 0.77$ $\sum w_{ei} R_{ei LP} = -0.79$	$R_{c HP} = 0.91$ $\sum w_{ei} R_{ei HP} = -0.24$
Correlation 1	$K_{eff LP} = 0.12$ $R_{LP}^2 = 0.42$	$K_{eff HP} = 0.52$ $R_{HP}^2 = 0.74$
Correlation 2	$R_{\%} = 0.75$ $u_{thd} = 0.50m/s^2$	$R_{\%} = 0.89$ $u_{thd} = 0.20m/s^2$
Correlation 3	$R_n = 0.24$	$R_n = 0.49$
Score = 4.8		

6.2 Validation with professional drivers

Three professional drivers participated in the MCA performance evaluation. The first driver has experience testing various high-performance driving simulators worldwide. The second driver usually suffers from motion sickness in our previous driving simulator ULTIMATE. The last driver had no prior experience with driving simulators before testing ROADS.

The optimization of MCAs parameters was facilitated by a co-author with expertise in vehicle dynamics mapping.

None of the three professional drivers experienced motion sickness during ROADS driving tests. Although the third driver initially found the car's feeling unnatural compared to a physical car in CTA1 scenario (first driving experience), he quickly adapted to using ROADS to evaluate the performances of various vehicles without issue. Their oral feedback regarding the representativeness of the physical car correlated with the scores presented in this paper.

For the N104 scenario, professional drivers could feel the Renault car's characteristics with MPC-MCA, while only sensing the tendencies of the car's behaviour with CF-MCA, if the tire model or anti-roll bar's diameter were modified. Regarding the CTA1 scenario, CF-MCA's performance was assessed by driving simulator team and deemed much less performant compared to MPC-MCA. Due to the workload on the ROADS, it is not yet evaluated by the professional driver.

The longitudinal acceleration is also a useful indicator to help the driver to better control the vehicle in high dynamic driving experience (CTA1 scenario). It is tuned to have the best performance below 2m/s^2 .

During the driving experiences, we changed the graphic engine in SCANer from OSG to Unreal. A significant amelioration has been observed in terms of visual feeling and global feeling by the professional drivers.

7. Conclusion

By using ROADS and MPC-MCA, we have developed a method to assess the high dynamic behaviours on handling track of a passenger car. The method is validated by professional drivers and applicable for vehicle project development.

Thanks to the performance of MPC-MCA in reproducing low frequency motion, the reproduction of high-frequency motion is also improved compared to CF-MCA, thus, increasing the performance of ROADS.

A numerical method to evaluate the performance of MCA is proposed. It will help us to improve again the achieved global score. The method can also be used for the performance evaluation of ROADS driving simulator.

In future, we will integrate the driving simulator's latency to reach the best performance of ROADS.

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